

SECTION 1: OS Layer Identification:

The Scenario mainly involves the application layer, runtime environment, application framework, Security Services, kernel and hardware.

* **Android**: Linux kernel + Android Runtime (ART) + Application Framework.

* **iOS**: XNU / Darwin kernel + iOS runtime / frameworks.

* Both use application Sandboxing, permission management, and hardware access Controls.

SECTION 2: Layer - by - Layer Technical Analysis.

When an application requests Contacts, Camera, Storage or Location:

Application → Runtime / Framework →
 permission Manager → kernel / OS Service
 → Hardware / Data → Application.

The OS checks whether the application has the required permission. If granted, the relevant System Service accesses the resource. Sandboxing prevents the application from directly accessing other applications' data.

Android generally uses a more user-visible runtime permission model, while iOS uses strict Sandboxing and controlled APIs with permission prompts.

SECTION 3: Root Cause & Trade-off Diagnosis.

The main security concern is unauthorized access to sensitive resources if permissions are overly broad or poorly controlled.

Sandboxing and permission checks improve security but add overhead and restrictions. Stronger isolation can reduce compatibility and flexibility,

while greater application access can improve functionality but increase security and privacy risks.

Android Security Trade-offs

Android provides considerable flexibility across many manufacturers and device configurations. Its permission model has evolved toward more granular runtime permissions and restrictions such as Scoped Storage.

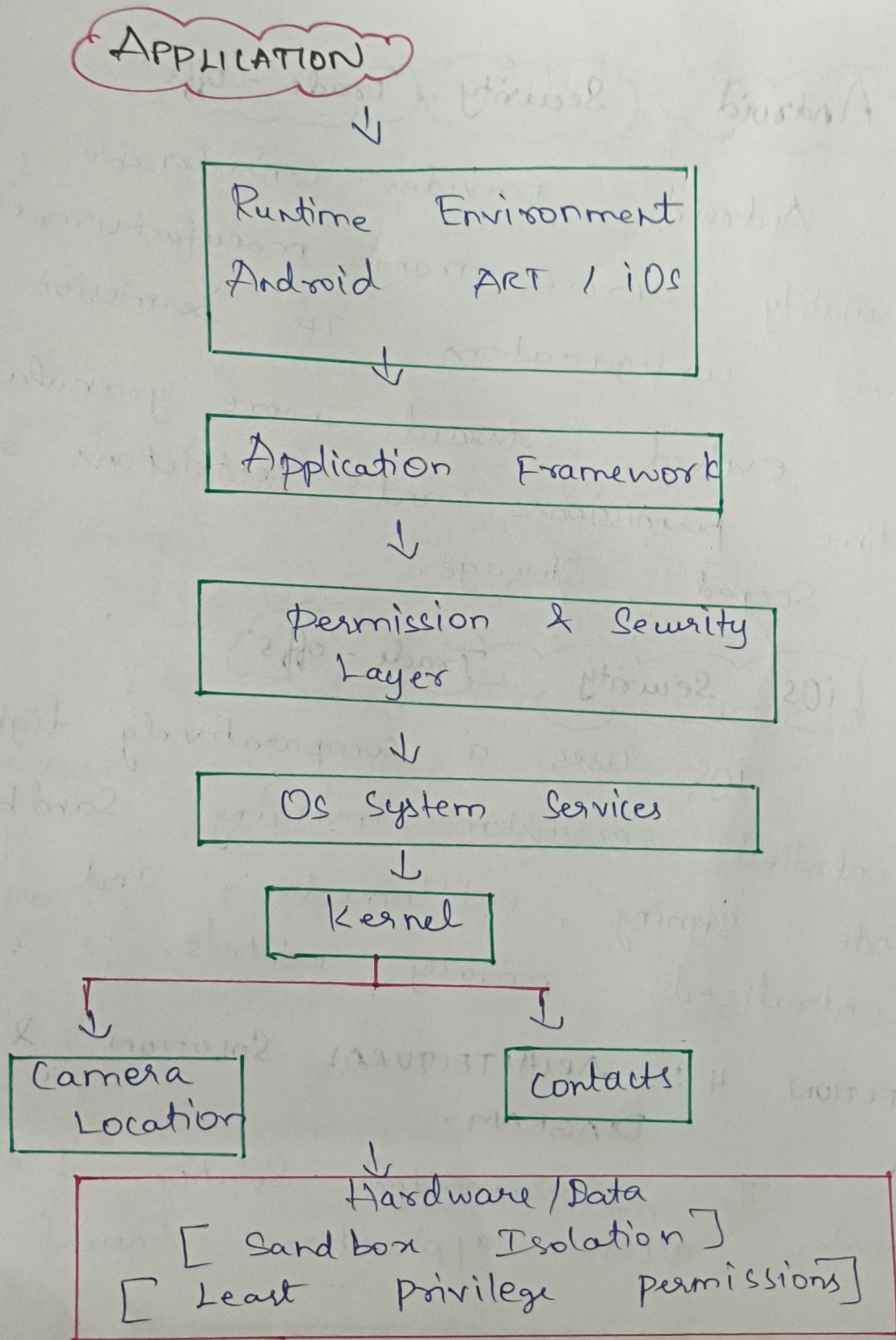
iOS Security Trade-offs

iOS uses a comparatively tightly controlled ecosystem involving Sandboxing, Code Signing, entitlements, and Centralized privacy controls.

SECTION 4 : ARCHITECTURAL SOLUTION & DIAGRAM.

The recommended solution is least-privilege access: applications should request

only the permissions they actually need, with runtime permission checks and Sandbox enforcement.



CONCLUSION:

Android and iOS both protect applications through Sandboxing and permissions, but their architectures and implementation differ. Combining Sandboxing, Least - privilege permissions, runtime checks, and Secure OS services provide stronger application security while maintaining functionality.